Andum — current state

Packages for money and assets · backend, worker, NEAR + EVM contracts, landing page
Report date: 12 Nov 2026 · everything below was run locally during this session

288	131	38
TypeScript tests (79 shared · 146 backend · 63 worker)	Rust tests (124 unit · 7 wasm sandbox)	Solidity tests

1. Summary

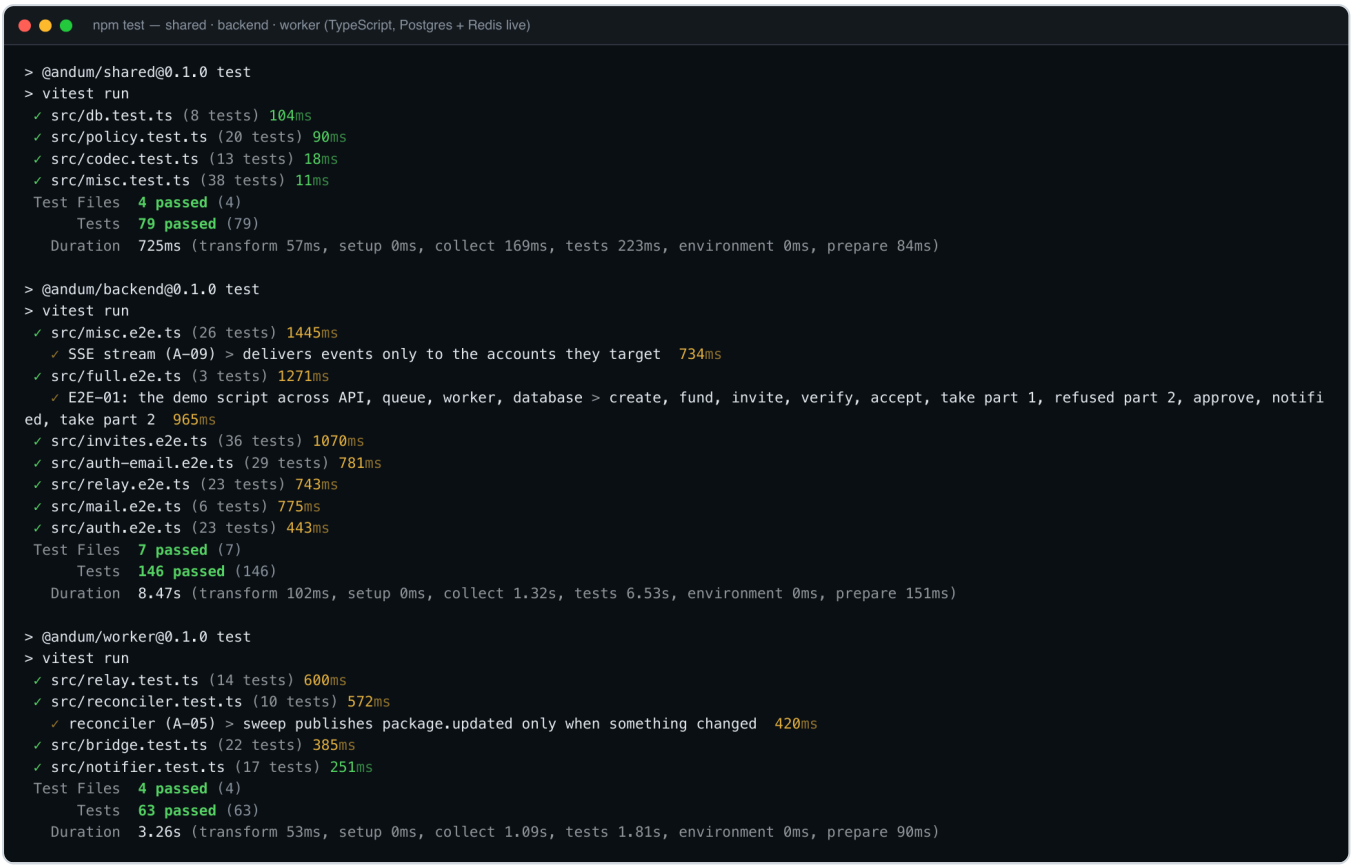
The backend (NestJS API, BullMQ worker with relayer, reconciler, notifier and bridge), the NEAR contracts (core, mock token, bridge) and the EVM contracts (mock token, gateway) are built and all automated suites pass in a fresh run today. Email sign-in over SMTP (non-custodial) is in place. A narrated end-to-end demo drives the real API, Redis queue, worker and Postgres through the full flow: login, create, fund, invite, accept, claim, approval and notification, including the refusal cases.

Read this before trusting the demo. Nothing is deployed. The demo and the full-stack tests use an **in-memory model of the core contract** behind a fake chain; the real Rust contract is proven separately by its own unit tests and by 7 sandbox tests on the compiled wasm. The relayer's real NEAR RPC client has not run against testnet, and Privy's NEAR support is unverified. The landing page (`andum-web`) is a separate project and is not connected to this backend yet. There is no product web app UI to screenshot, so the UI screenshots are of the landing page only.

2. Test results (fresh run)

Area	Command	Result
shared library	`npm run test -w shared`	**79 passed**
backend API (incl. 3 full-stack API+queue+worker scenarios)	`npm run test -w backend`	**146 passed**
worker (relayer, reconciler, notifier, bridge)	`npm run test -w worker`	**63 passed**
NEAR core contract	`cargo test -p andum-core`	**102 passed**
NEAR mock token / bridge	`cargo test --workspace`	**13 + 9 passed**
NEAR sandbox (real wasm, cross-contract calls)	`cargo test -p andum-integration`	**7 passed**
EVM (Foundry)	`forge test`	**38 passed**

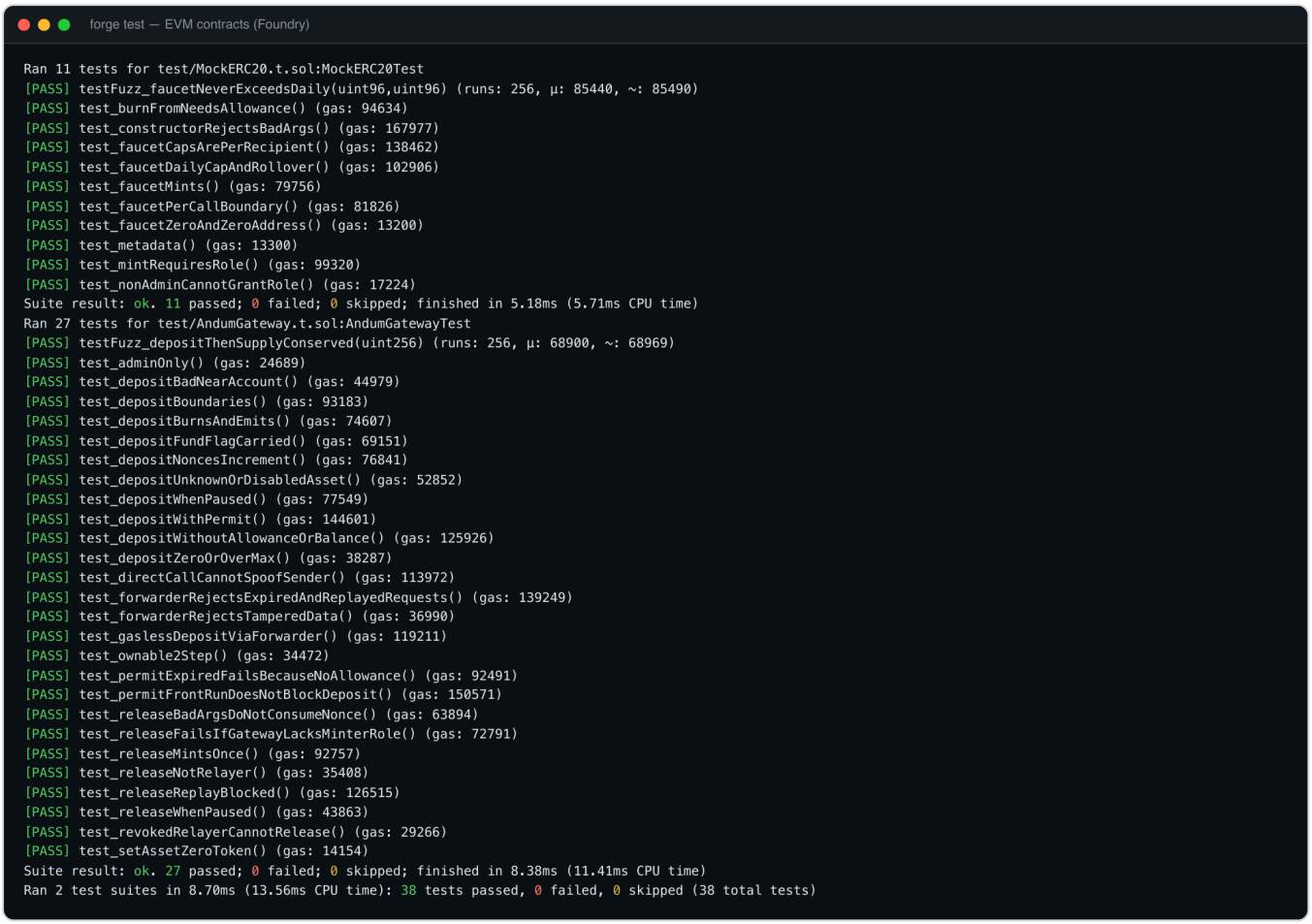
Total: 288 TypeScript + 131 Rust + 38 Solidity = 457 automated tests, 0 failing. Tests cover positive, negative and edge cases (wrong secrets, replays, double claims, expiry boundaries, rate limits, header injection, SMTP outages, policy denials).



TypeScript suites against live Postgres and Redis



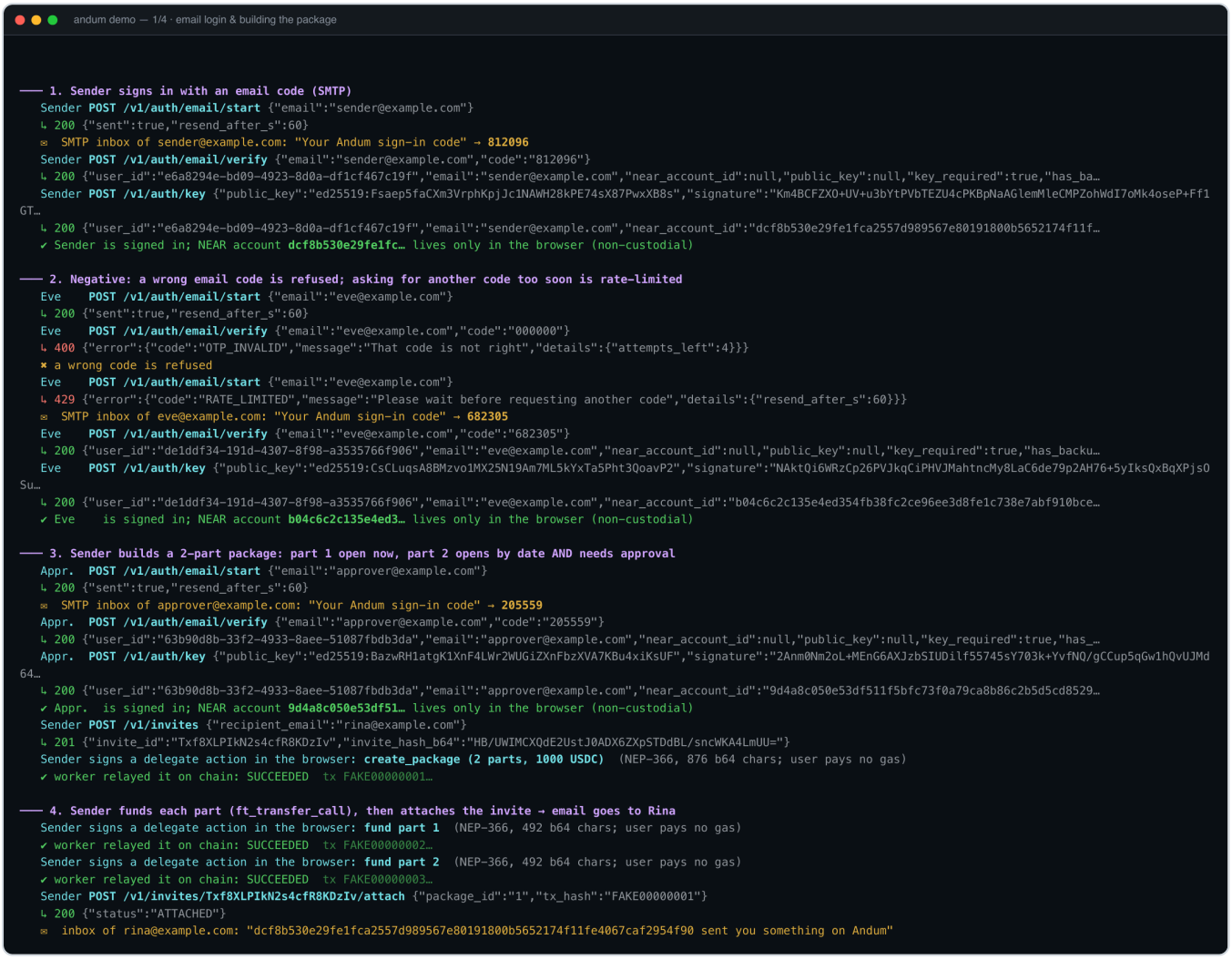
NEAR contract tests (first 48 lines; full run has 131 tests)



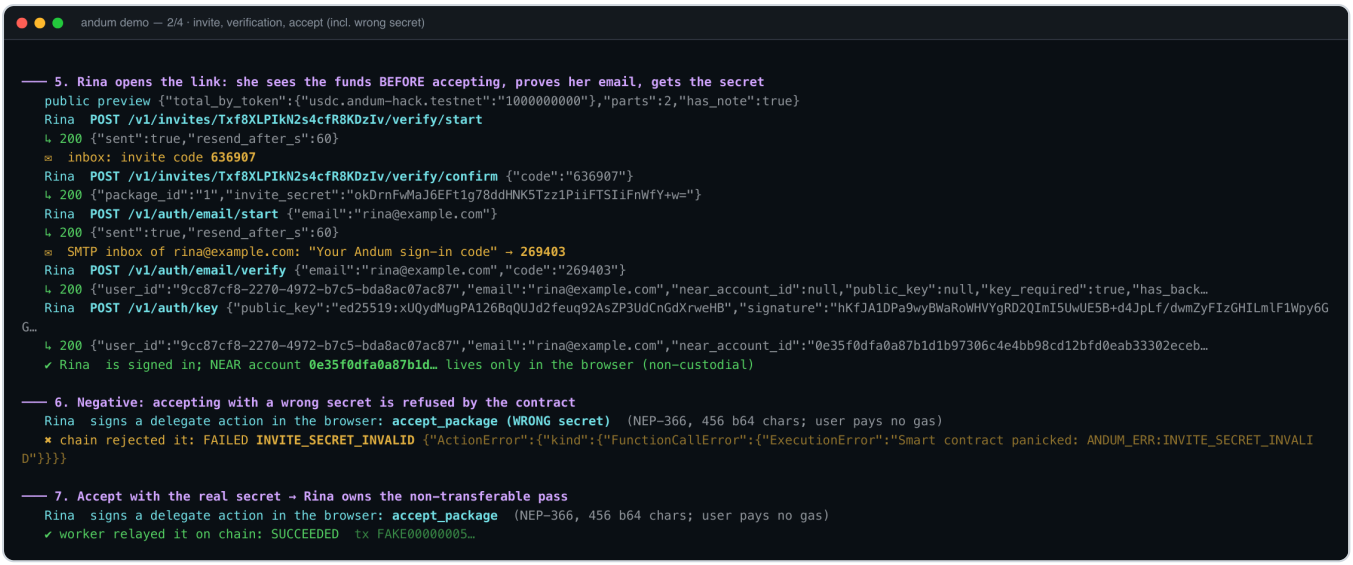
Foundry tests

3. End-to-end backend demo (narrated)

Recorded as `demo-backend-flow.mp4`. Script: `scripts/demo/story.demo.ts`. Every request below is a real HTTP call to the NestJS app; relays go through BullMQ, are processed by the real worker, and land in Postgres. Only the chain is simulated.



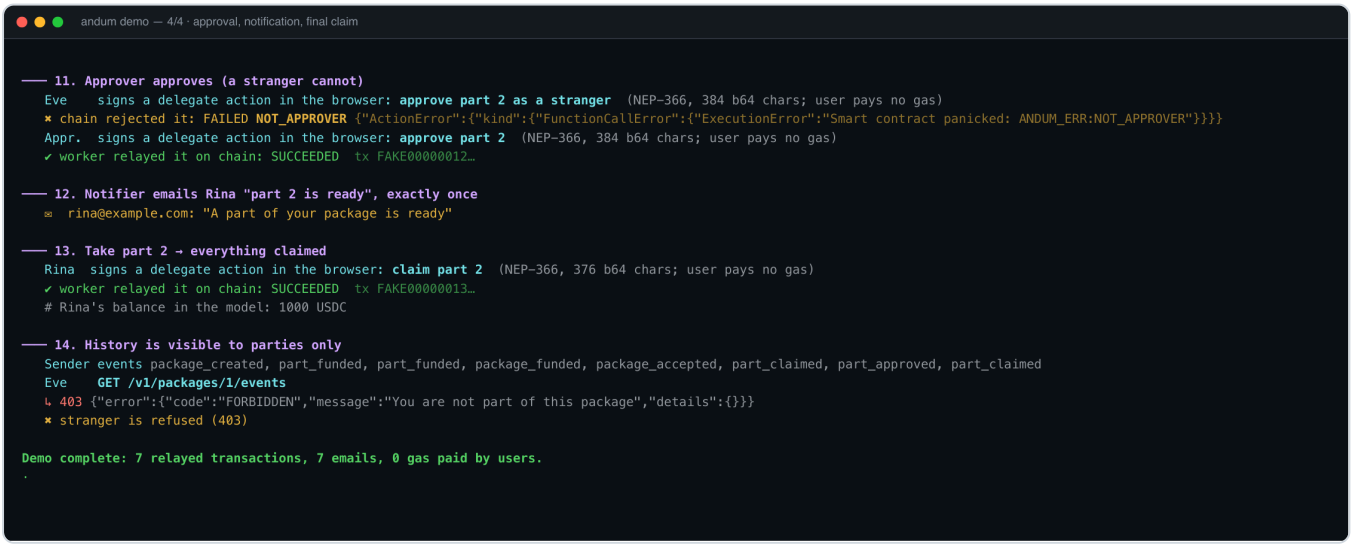
Steps 1–4: email code login over SMTP, a wrong code refused, a resend rate-limited, package created and funded through gasless delegate actions



Steps 5–7: the receiver sees the funds before accepting, verifies by email, a wrong invite secret is rejected with `INVITE_SECRET_INVALID`, the real secret succeeds



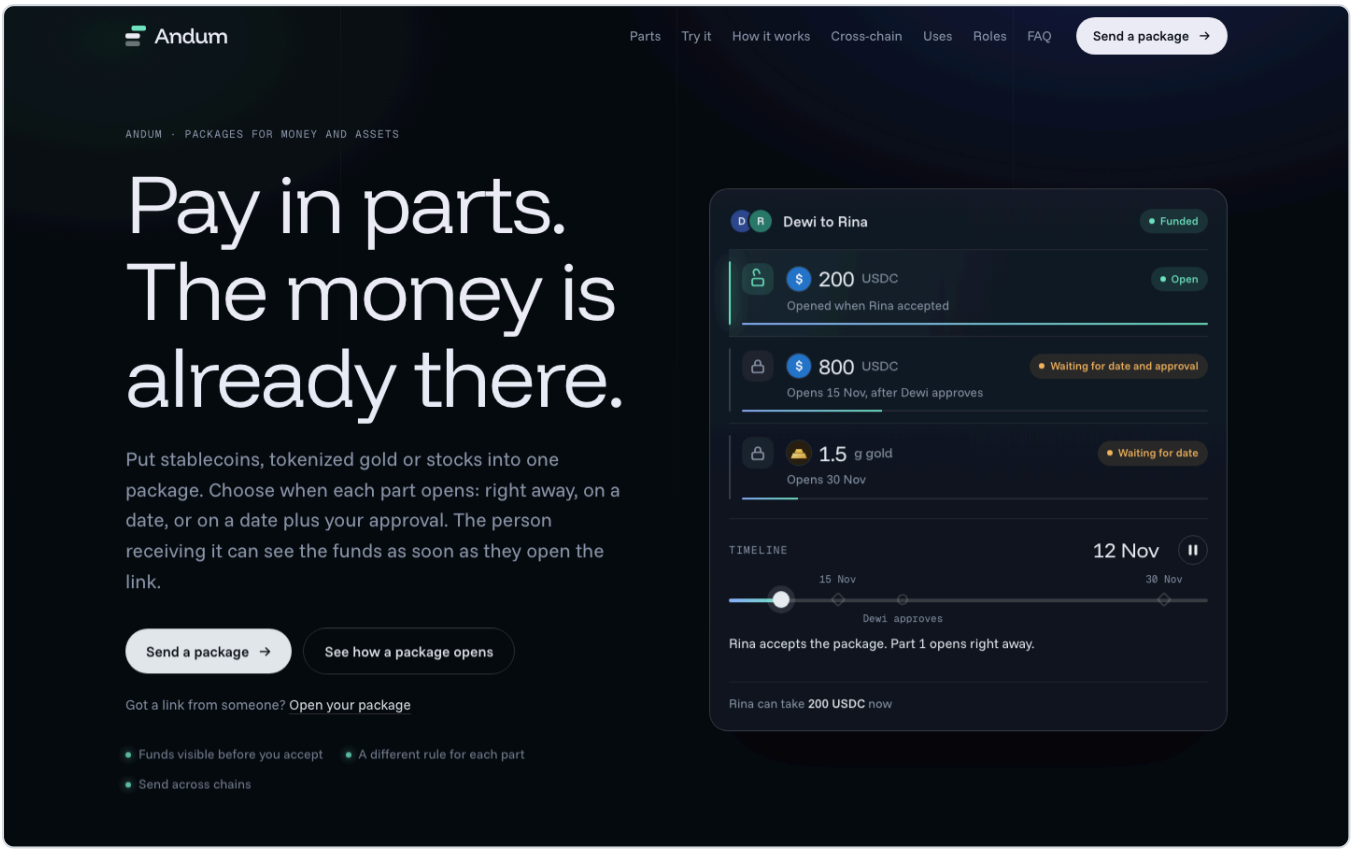
Steps 8–10: part 1 claimed; too early, twice, by a stranger, and via a non-allowlisted contract are all refused; approval still missing after the date



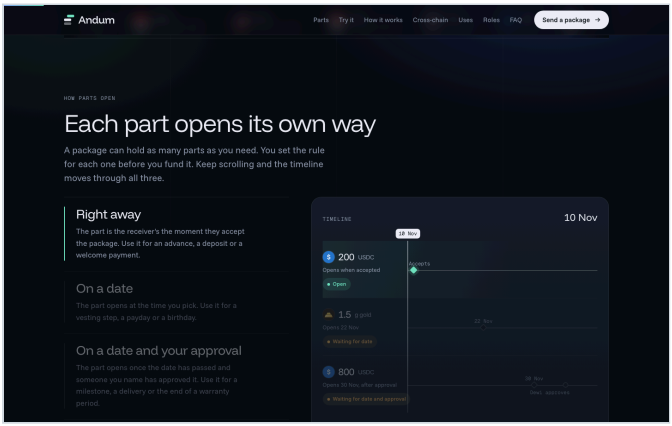
Steps 11–14: stranger cannot approve, approver can; "part ready" email sent once; final claim; history visible to parties only

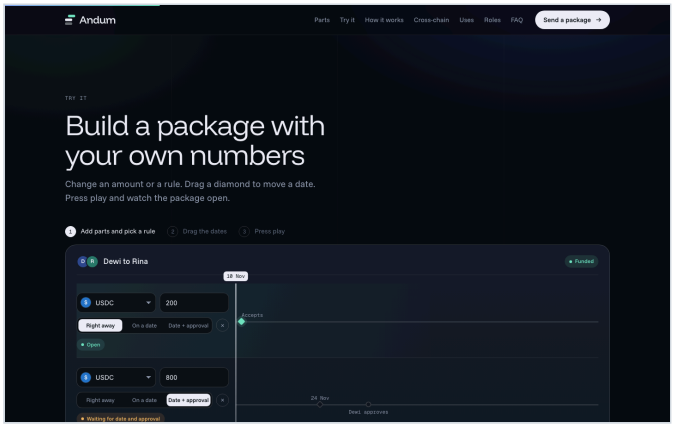
4. Landing page (andum-web)

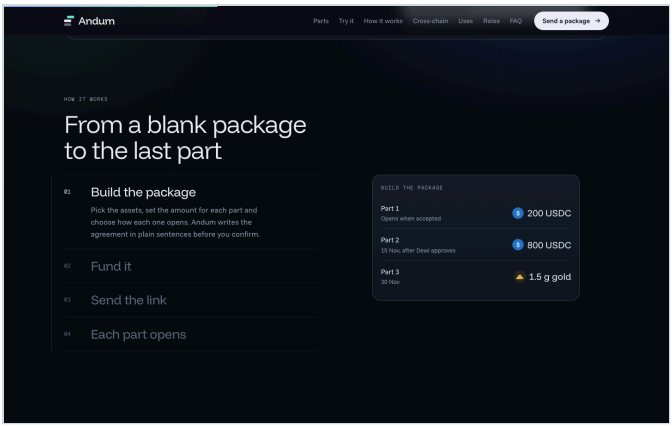
Vite + React + Tailwind page with a receiver preview route. Recorded as `demo-landing-page.mp4` (sped up 3×). This was built separately; it is static and not wired to the API.

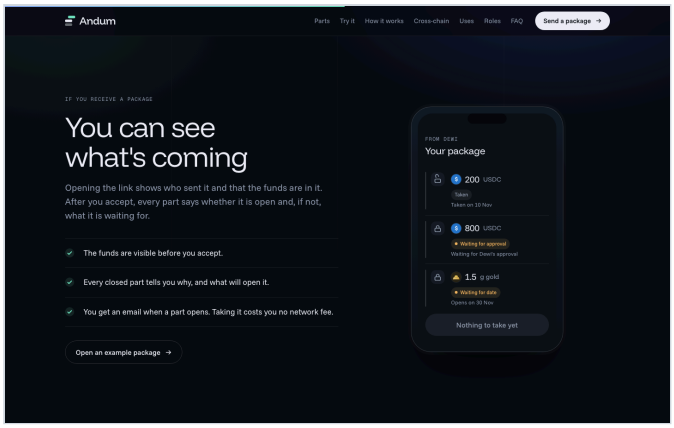


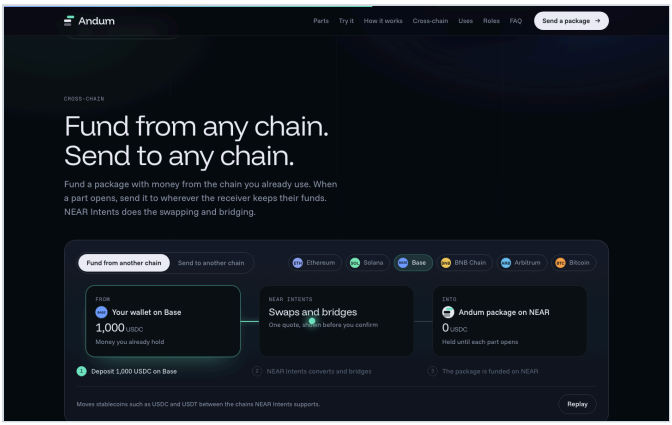
Landing page, top of page (desktop 1440 px)

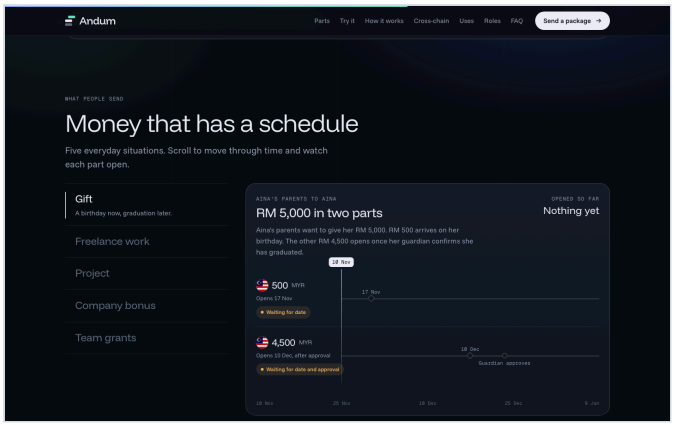




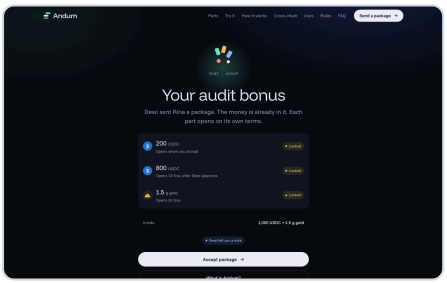




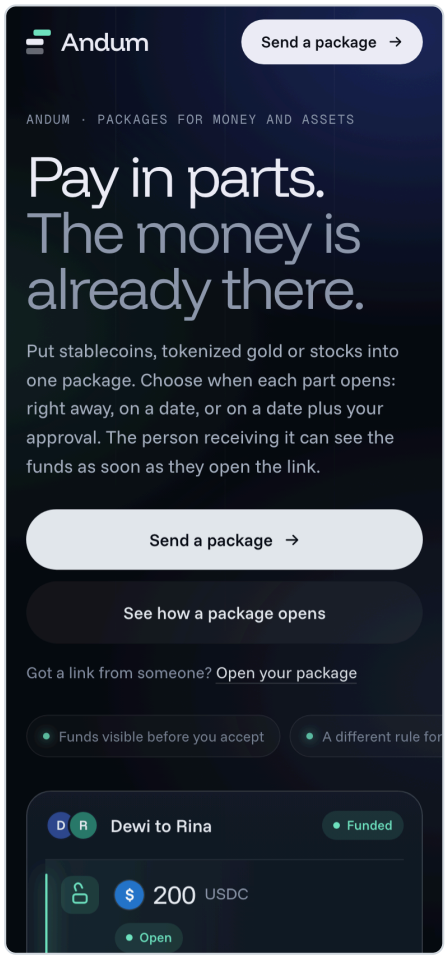




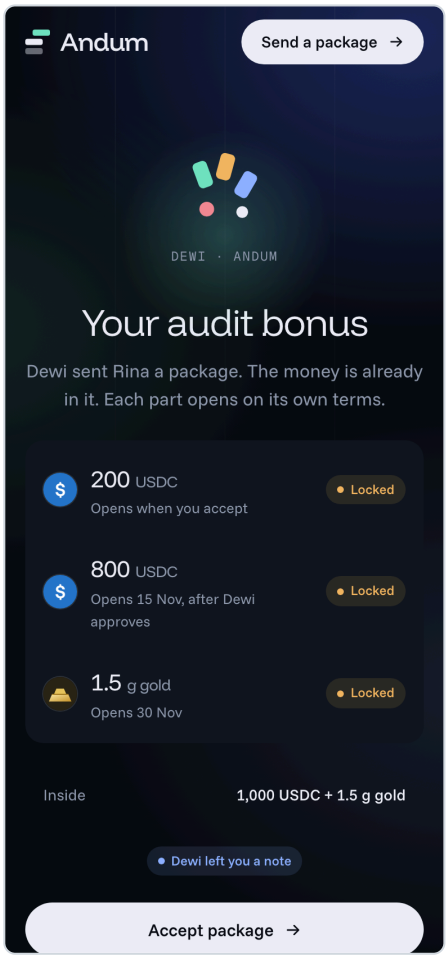
Sections of the full page, in order (full-page capture: `web-landing-full.png`).



/p/rina-audit-bonus



Mobile, landing



Mobile, receiver link

5. What is done and what is not

Area	State	Notes
NEAR core contract (create, fund, accept, approve, claim, reclaim, NFT views, events)	**Done**	Unit + wasm sandbox tested. Wasm must be built with `--cfg near` (configured).
Mock token, faucet, NEAR bridge contract	**Done**	Bridge fund path unit-tested only
EVM mock token, gateway, ERC-2771 forwarder	**Done**	Foundry tested
Backend API (auth, invites, packages, relay, faucet, notifications, SSE)	**Done**	EVM faucet endpoint returns 501
Email login with SMTP, client-held key, encrypted backup	**Done**	Tested against a real in-process SMTP server. Real provider + SPF/DKIM needed.
Privy login	**Partial**	Token verification built; NEAR signing through Privy unverified
Relayer, policy engine, reconciler, notifier	**Partial**	Works against fake chain; real RPC client never run on testnet
Bridge worker	**Partial**	State machine tested with fakes
Deployment to NEAR testnet / Sepolia / Base Sepolia	**Not done**	Needs funded accounts and keys (human)
Product web app wired to the API, Playwright E2E, UAT	**Not done**	Landing page exists separately
OpenAPI file, `og.png` renderer, runbook, decisions log	**Not done**	
Docker compose	**Not run**	Docker was not running

6. Recommended next steps

1. Create and fund the NEAR testnet root account, run `scripts/deploy-near.sh`, then spikes S-01 (gasless `ft_transfer_call` with 1 yocto) and S-03 (gas numbers).
2. Validate the hand-rolled RPC client against testnet; if it misbehaves, swap in `near-api-js` behind the same interface.
3. Wire the web app to the API (login, create, receive) and add Playwright E2E.
4. Choose an SMTP provider and verify the sender domain; run spike S-02 for Privy.

7. Files delivered

File	Content
andum-report.pdf	this report

demo-backend-flow.mp4 (≈43 s)	narrated end-to-end backend flow
demo-test-suites.mp4 (≈13 s)	all test suites passing
demo-landing-page.mp4 (≈73 s)	landing page and receiver link walk-through
screenshots/*.png	every image used above